Remaining Useful Life Estimation and Health Monitoring of EV Battery Systems Using Deep Learning Models

1st Tanisha Gangrade
ECE Department
MANIT
Bhopal, India
tanishagangrade29@gmail.com

2nd Lalitha Anoojna Paturi
ECE Department
MANIT
Bhopal, India
anoojnapaturi42@gmail.com

3rd Dr. Suryakant Gautam
ECE Department
MANIT
Bhopal, India
suryakant@manit.ac.in

***Abstract*—In recent times, concerns for environment safety have increased and sustainable options like Electric Vehicles (EVs) are preferred over transportation that uses fossil fuels. However Lithium-ion batteries slowly lose their initial capacity, with every charge and discharge cycle and it is important to know how much longer the battery can be safely used. This study uses NASA's Lithium-ion battery dataset, which contains operational data collected from several batteries tested until their capacity reached the failure threshold. This paper proposed a hybrid CNN-LSTM architecture through which we are able to analyze the structural patterns as well as the temporal changes in the data as the aging of the battery increases. The performance of the proposed approach is evaluated using error metrics such as Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). The results show that the model is able to make the predictions with low error values for various batteries with different starting points. The model also performs well when predicting both earlier and later stages of the battery degradation. Hence, hybrid deep learning models are promising tools for battery prognostics and may be readily integrated into BMS for predictive maintenance to ensure safe and efficient battery operation in EVs.**

***Index Terms*—Remaining Useful Life, Deep Learning, Battery Management System, Electric Vehicles, NASA Lithium-ion battery set**

I. INTRODUCTION

Modernization has rapidly increased the demand on the transportation systems, bringing forward some of the major issues which affect both individual and societal well being as a whole such as the growing health issues caused by fuel inefficiencies [1], [2]. Recent studies show that every year vehicular pollution contributes to nearly two-thirds of urban air pollution, emitting harmful gases such as carbon monoxide (CO), nitrogen oxides (NOx), and hydrocarbons, which severely affect human health and the environment [1]. Additionally, traffic congestion in urban areas causes more fuel consumption, which leads to economic losses, and inefficiencies in the transportation system, especially in India, where heterogeneous traffic conditions further worsens the scenario [2].

To deal with these challenges the Electric Vehicles (EVs) comes out to be a solution due to their promising results in reducing the environmental pollution and dependencies on fossil fuels. However, despite having a strong government support, incentives and policies, the adoption of EVs continues to be minimal due to several barriers such as inadequate charging infrastructure and limited driving range. Simultaneously, the primary motivation for consumers to go for the adoption of EV is due to their reduced emissions and environmental benefits, underlining the need for technological advancements that enhances the reliability of EVs and their performance [3].

An important factor influencing the performance of EVs is the lithium-ion battery system, which decides the efficiency of the vehicle, its safety, and the life of its operation. Over time, the degradation of batteries has led to a decrease in capacity and the failure of potential systems, marking a need for correct monitoring of the battery health. Remaining Useful Life (RUL), defined as the time duration from the current state to the end of operational life, serves as an important indicator for predictive maintenance and system reliability [4].

Traditional methods for the estimation of RUL can be categorized into physics based, experimental and data driven methods [4], [5]. While the physics based models have the capability for estimation with high accuracy, they require the knowledge of electrochemical processes in much detail and thus the computational cost for these methods are high [6]. Furthermore, the data driven methods depend more on the historical data to model the degradation trends and are more suitable in the scenarios where large datasets are available [4]. Machine Learning (ML) techniques such as Support Vector Regression (SVR) are being employed to directly estimate the RUL from sensor data, thus improving the computational efficiency and reducing the model complexity [7].

The classical regression models are considered a baseline for the estimation of RUL but they assume a linear relationship between the input features and the battery degradation behaviour, which is not consistent with the aging of the battery being non linear. Ensemble methods like the Random Forest improves the predictive capabilities by modeling the non linear relationships through decision trees [8]. However, since the model takes each observation independently, it fails to identify the temporal dependencies [9]. Moreover, gradient boosting methods such as XGBoost in particular improves the prediction accuracy by sequentially minimizing the residual errors and efficiently captures the model complex feature interactions [10].

Nevertheless, conventional ML models at times fail to capture the temporal dependencies that are being inherited in battery degradation processes. Artificial Neural Networks (ANNs) and Recurrent Neural Networks (RNNs) have been explored for modeling the time series data and they are being effective in estimating the RUL respectively [11], [12]. However, standard ANNs and RNNs have limitations in learning long term dependencies. Therefore, Long Short-Term Memory (LSTM) has been proposed to improve the capability in capturing the long-term temporal relationships and extract the hidden degradation patterns from the sequential data. LSTM-based models outperformed the traditional regression and convolution based approaches [13].

Recently deep learning architectures such as the combination of Convolution Neural Network (CNN) and LSTM surpassed the conventional methods by jointly capturing the local feature patterns and temporal dependencies [14]. CNN layers act as feature extractors, they identify the short-term fluctuations and local degradation patterns, while LSTM layers model the long term sequential behaviour. We will discuss the approach in the coming section.

II. Dataset Description & Analysis

A. Dataset Description

The dataset used in this paper from the NASA Prognostics Center of Excellence lithium-ion battery repository, consists of the degradation of lithium-ion batteries that is collected under the controlled experimental environment until they reach the End-of-Life (EOL). The dataset contains the repeated charge-discharge cycles of multiple batteries. Here we selected batteries B0005, B0006, B0007, and B0018 because of their diverse degradation characteristics and frequent use in predicting the RUL [15].

TABLE I
SUMMARY OF BATTERY TESTS AND EXTRACTED FEATURES

UID	Type	Temp (°C)	Battery ID	Test ID	Capacity (Ah)	Re (Ω)	Rct (Ω)
1	Discharge	4	B0047	0	1.6743	–	–
2	Impedance	24	B0047	1	–	0.05606	0.20097
3	Charge	4	B0047	2	–	–	–
4	Impedance	24	B0047	3	–	0.05319	0.16473
5	Discharge	4	B0047	4	1.5244	–	–

[a] UID: unique record identifier; Type: Charge, Discharge, or Impedance; Temp: ambient temperature; Battery ID: battery identifier; Test ID: test sequence; Capacity: discharge capacity (Ah); Re: electrolyte resistance (Ω); Rct: charge transfer resistance (Ω); "–" indicates unavailable data.

Among these, the discharged cycles are mainly used for the degradation analysis, as the capacity measurements are only available during the discharge operations. Additional understanding can be derived from the impedance measurements (Re, Rct) providing internal resistance growth but are sparsely available and are not aligned with every cycle.

The dataset is highly sequential, that is each observation depends on the previous degradation states. Features such as capacity, voltage, current and temperature evolve over cycles, forming a multi-varying time series data [15].

The batteries are bound to repeat the cycles until their capacity drops below a predefined failure threshold. In this paper, we set the failure threshold to 70% of the rated capacity.

Battery storage systems go through gradual degradation of the capacities due to aging mechanisms such as electrochemical reactions and thermal effects. As highlighted in paper [16], effective data analysis and prognostics are important for the management of the lifecycle of the batteries and ensuring the system reliability. Moreover, data-driven approaches utilise historical sensor data to model the degradation without requiring explicit physical modeling [15], [16].

B. Data Analysis

In this paper, a complete Exploratory Data Analysis (EDA) is conducted to characterize the battery degradation behaviour, identity failure relationships and prepare the dataset for predictive modeling. The analysis primarily focuses on the discharge cycle data, as the capacity measurements are only available during the discharge operations and directly reflect the health of the batteries.

B.1 Preprocessing

The raw dataset has mixed operational records such as charge, discharge and impedance. For the degradation modeling we only require the discharge cycles and they are ordered sequentially to form a cycle based time series:

$$\mathcal{D} = \{(cycle_i, C_i)\}_{i=1}^{N} \tag{1}$$

where:

- $cycle_i$ denotes the cycle index
- C_i denotes the discharge capacity at cycle i

B.2 State of Health (SOH) Normalization

Since the absolute capacity values vary across multiple batteries, normalization is required to allow cross-battery comparison and stable model training. The SOH is calculated as:

$$SOH_i = \frac{C_i}{C_0} \tag{2}$$

where:

- C_0 is the initial capacity of the battery
- C_i is the capacity at cycle i

Due to the normalization all the battery trends are bound to be in a common range between 0 and 1 with SOH being 1 means battery is in its best health and SOH being nearly 0.7

implies the EOL threshold. This is a standard form with battery prognostics practices [15].

B.3 Feature Normalization & Scaling

For the enhancement of the model and for better convergence we scale and normalize the features, this helps prevent the dominance of large scale features, input variables such as capacity and cycle index. This makes sure that all the features are bound to follow the range, improving training stability especially for neural networks. We applied the Min-Max normalization technique:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{3}$$

where:

- x is the original value
- $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of the feature
- x' is the normalized value

B.4 Capacity Degradation

Batteries B0005, B0006, B0007 have gradual and smooth decay whereas battery B0018 has accelerated degradation.

This variation underlines the presence of distribution shift, which is a key challenge in predicting the RUL. The degradation behavior follows three distinct phases:

1. Initial Phase - Having Slow degradation due to stabilization of electrochemical processes.
2. Mid-Life Phase - Has approximately linear degradation.
3. End-of-Life Phase - Has abrupt capacity drop due to accelerated aging.

This nonlinear behavior confirms that linear models are insufficient for accurate prediction of the RUL.

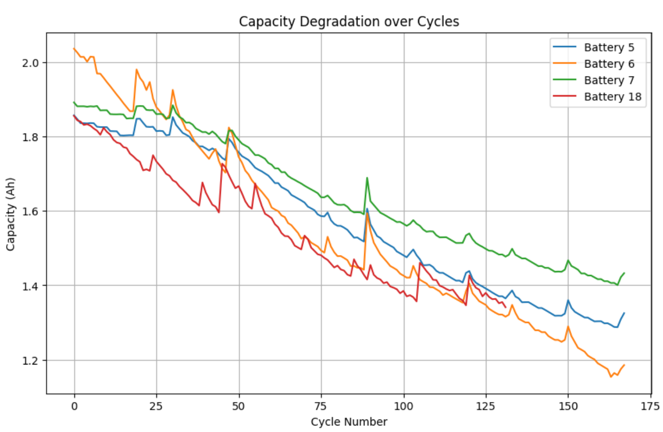

Fig. 1. Capacity degradation over cycles for batteries B0005, B0006, B0007, and B0018.

B.5 SOH Degradation Trends

- All batteries starts at SOH=1
- SOH decreases monotonically with cycles
- The slope of degradation varies across batteries

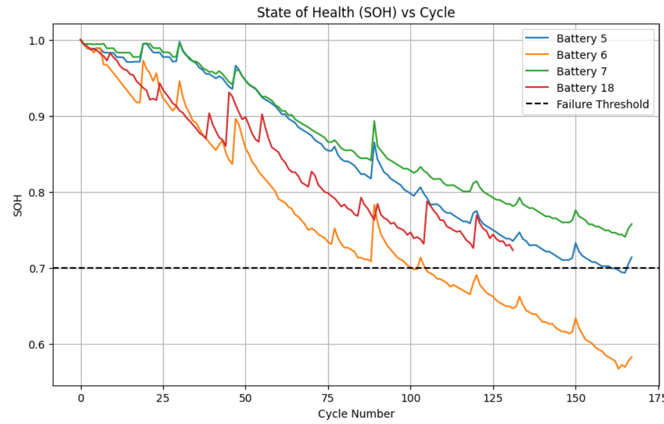

Fig. 2. State of Health degradation over cycles for batteries B0005, B0006, B0007, and B0018.

The normalized SOH representation allows direct comparison across batteries, which is not possible using raw capacity values. Additionally, the SOH plot also highlights the reduced variance due to normalization.

B.6 Threshold Based Failure Analysis

A failure threshold is defined as:

$$SOH_{\text{threshold}} = 0.7 \tag{4}$$

Batteries are typically considered at EOL region when their capacity drops to 70% of the original capacity, implying it has lost 30% of its usable capacity. Therefore, SOH threshold is taken as 0.7. This 70% of the initial capacity is commonly used as the EOL criterion in battery prognostics [15].

The intersection point of the SOH curve with this threshold determines the RUL target:

$$RUL = cycle_{EOL} - cycle_{current} \tag{5}$$

The following observations therefore indicate that the traditional regression based methodologies are insufficient in capturing the full dynamics of the battery degradation. We will further discuss the methodology in the coming section.

III. Proposed Methodology

This paper proposes a hybrid CNN-LSTM framework that can address the limitations of traditional ML and deep learning models together. This solution can effectively model the non linear sequential lithium-ion battery degradation trends for the prediction of SOH and RUL estimation. The raw data is preprocessed by filtering the discharged cycles and constructing the cycle-wise degradation sequences, followed by normalization of the computed SOH through data analysis. To learn temporal dependencies over sequences of fixed lengths, the sequential data are further converted into samples via a sliding window approach, and the model learns in a supervised manner.

The proposed architecture consists of one-dimensional convolutional layers, LSTM layers, and fully connected dense layers. To extract the local degradation pattern such as the short term variations and sudden capacity drops, the CNN layers apply the convolution filters over the input sequences, which most of the time are overlooked by the purely sequential models. These extracted features are then passed on to the LSTM layers, to capture the long-term dependencies and slow degradation trends across cycles using the gated memory techniques. At the final stage, the dense layers map the learned trends to SOH predictions. The model is trained using the Adam optimizer with mean squared error (MSE) as the loss function, ensuring stable convergence.

Compared to traditional ML models like ANN, Random Forest, and XGBoost, which fails to capture the temporal dependencies and standalone sequence model such as RNN and LSTM, which may not be effective in identifying the local variations, the proposed CNN–LSTM architecture provides a combined method that learns and both local variations and global degradations at the same time. This results in reduced error and better generalization across the batteries. The predicted SOH trend is further used in estimating the RUL by identifying the cycle at which the SOH crosses a predefined failure threshold, making the framework promising in real time health monitoring applications.

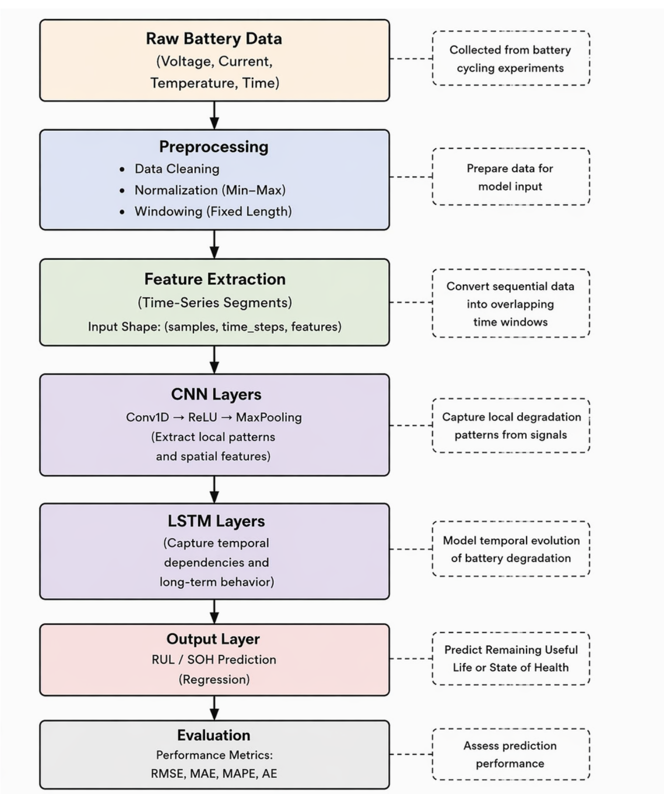

Fig. 3. Proposed CNN–LSTM Framework for the prediction of RUL in Lithium-ion Batteries

Algorithm 1 Detailed pseudocode for the CNN–LSTM Framework

Require: Battery dataset D
Ensure: Predicted Remaining Useful Life (RUL)

1: Load dataset D
2: **Data Preprocessing:**
  Handle missing values (if any)
  Select relevant features
  Normalize features using Min-Max scaling:
    $X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$
3: **Generate Time-Series Sequences:**
  Define window size W
  for $i = 0$ to $(N - W)$ do
    $X_{\text{seq}}[i] \leftarrow X_{\text{scaled}}[i : i + W]$
    $y_{\text{seq}}[i] \leftarrow RUL[i + W]$
  end for
4: **Split dataset:**
  Training set $\leftarrow 80\%$
  Testing set $\leftarrow 20\%$
5: **Reshape input for CNN–LSTM:**
  $X \rightarrow (\text{samples}, \text{time\_steps}, \text{features})$
6: **Initialize CNN–LSTM Model:**
  Add Conv1D layer (filters, kernel size)
  Apply ReLU activation
  Apply MaxPooling1D
  Add LSTM layer (units, return sequences)
  Add final LSTM layer
  Add output Dense layer (1 neuron)
7: **Compile model:**
  Loss function $\leftarrow$ Mean Squared Error (MSE)
  Optimizer $\leftarrow$ Adam
8: **Train model:**
  for $epoch = 1$ to E do
    Forward pass through CNN + LSTM
    Compute loss (MSE)
    Backpropagate error
    Update weights
  end for
9: **Predict on test data:**
  $y_{\text{pred}} \leftarrow model(X_{\text{test}})$
10: **Evaluate model:**
  MAE $\leftarrow \text{mean}(|y_{\text{true}} - y_{\text{pred}}|)$
  RMSE $\leftarrow \sqrt{\text{mean}((y_{\text{true}} - y_{\text{pred}})^2)}$
  MAPE $\leftarrow \text{mean}\left(\left|\frac{y_{\text{true}} - y_{\text{pred}}}{y_{\text{true}}}\right|\right)$
  AE $\leftarrow |y_{\text{true}} - y_{\text{pred}}|$
11: Return predicted RUL and evaluation metrics

IV. Results & Discussion

A. Graphical Results

The performance of our proposed CNN–LSTM framework is shown in the figure which gives the comparison between the

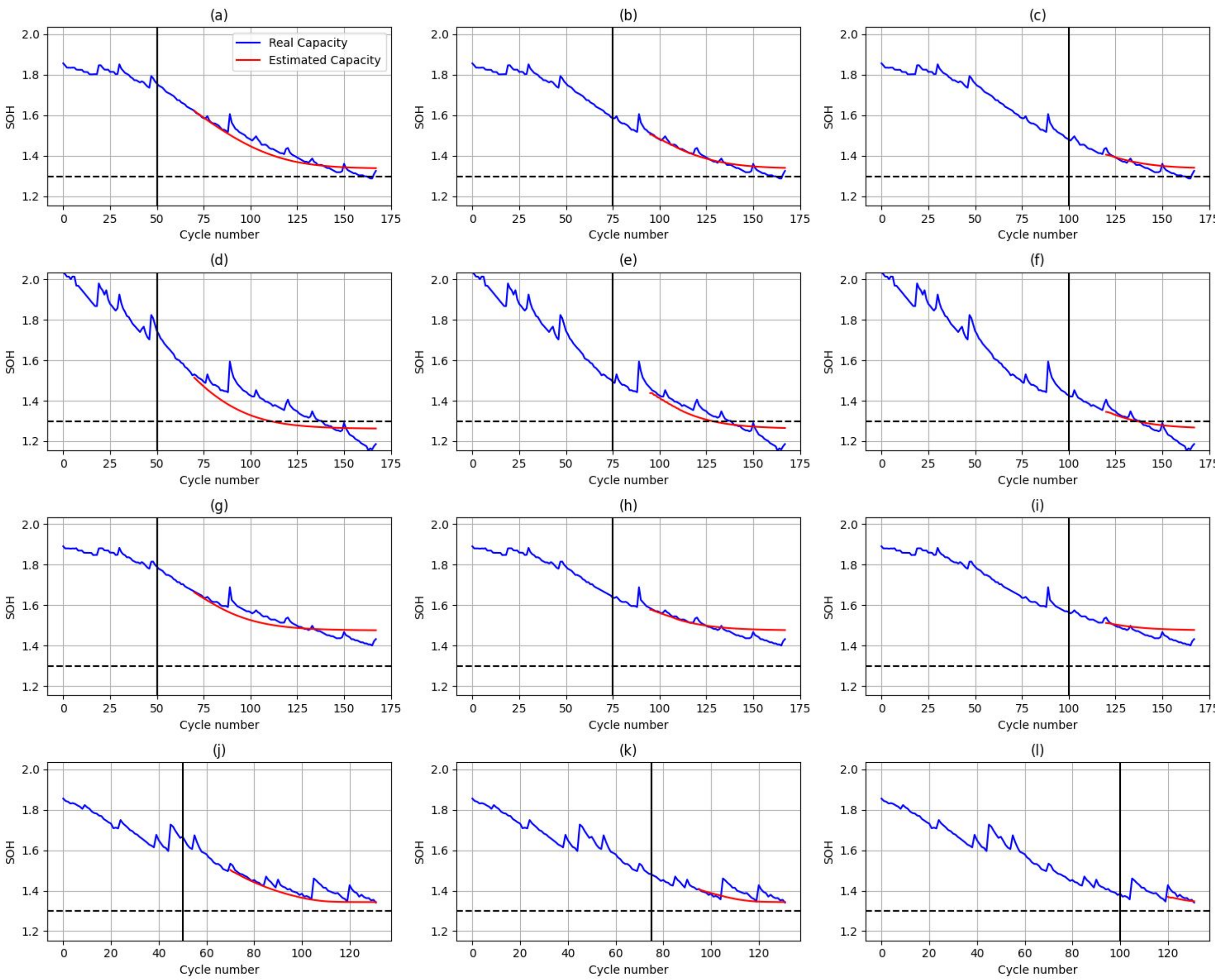

Fig. 4. Prediction results of four batteries at the starting points 50, 75, and 100: (a–c) for battery 5; (d–f) for battery 6; (g–i) for battery 7; and (j–l) for battery 18.

actual and predicted SOH degradation curves across batteries B0005, B0006, B0007 and B0018. The blue curves implies the true degradation curves while the red ones represent the proposed model predictions. The horizontal dashed line indicates the predefined EOL threshold of failure threshold. The vertical line in each one of the subplots implies the point after which the predictions are generated, indicating the capability of the model to predict future values based on the historical data.

It can be seen that the predicted SOH closely follows the overall degradation behaviour of the actual data, having both non-linear decay characteristics and long-term temporal dependencies. However small deviations are being present in the areas with abrupt fluctuations, and the predicted curves intersect the EOL threshold near the actual failure point, implying reliable estimation of the RUL.

In all, the results do show that our proposed hybrid CNN-LSTM framework efficiently models the battery degradation behaviours and provides long-term predictions, validating its suitability for real-world prognostics and health management applications.

B. *Quantitative Evaluation*

The performance of the proposed method, a hybrid CNN–LSTM framework, is evaluated and compared with multiple baseline models, which includes ANN, RNN and LSTM, using the NASA battery dataset for batteries B0005, B0006, B0007, B0018. The models are assessed using standard regression metrics, namely Mean Absolute Percentage Error (MAPE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).

B.1 Observation

1. Our proposed solution that is a hybrid CNN–LSTM model achieves consistently low error values across the bat-

TABLE II
PERFORMANCE COMPARISON OF ANN [11], RNN [12], LSTM [13] AND CNN–LSTM (PROPOSED) MODELS ACROSS DIFFERENT STARTING CYCLES

Start Cycle	**Model**	**MAPE**				**RMSE**				**MAE**			
		B5	B6	B7	B18	B5	B6	B7	B18	B5	B6	B7	B18
50	ANN	0.0742	0.0843	0.0891	0.0195	0.0593	0.0625	0.0770	0.0180	0.0555	0.0535	0.0700	0.0148
	RNN	0.1178	0.1725	0.0282	0.1085	0.0959	0.1256	0.0329	0.1089	0.0876	0.1098	0.0218	0.0813
	LSTM	0.0126	0.0338	0.0226	0.0312	0.0109	0.0301	0.0206	0.0266	0.0097	0.0214	0.0177	0.0234
	CNN–LSTM	0.0249	0.0293	0.0089	0.0182	0.0235	0.0257	0.0097	0.0194	0.0194	0.0186	0.0072	0.0139
75	ANN	0.0632	0.0914	0.0940	0.0242	0.0496	0.0662	0.0791	0.0226	0.0462	0.0567	0.0728	0.0183
	RNN	0.0424	0.1594	0.0400	0.1099	0.0333	0.1100	0.0345	0.1004	0.0313	0.0998	0.0310	0.0819
	LSTM	0.0093	0.0430	0.0241	0.0273	0.0077	0.0347	0.0217	0.0225	0.0068	0.0263	0.0185	0.0202
	CNN–LSTM	0.0120	0.0387	0.0099	0.0216	0.0106	0.0304	0.0093	0.0209	0.0090	0.0237	0.0077	0.0164
100	ANN	0.0468	0.0916	0.0980	0.0187	0.0359	0.0636	0.0794	0.0185	0.0335	0.0554	0.0748	0.0141
	RNN	0.0150	0.0462	0.0537	0.0458	0.0142	0.0343	0.0542	0.0352	0.0108	0.0278	0.0406	0.0339
	LSTM	0.0079	0.0534	0.0254	0.0173	0.0070	0.0396	0.0222	0.0150	0.0057	0.0321	0.0194	0.0128
	CNN–LSTM	0.0085	0.0452	0.0110	0.0139	0.0079	0.0340	0.0103	0.0133	0.0062	0.0272	0.0084	0.0104

teries B0005, B0006, B0007 and B0018. MAPE values are generally below 5% indicating good prediction accuracy.
2. Early vs Late Prediction Performance

- At start cycle 50 (early prediction): Errors are slightly higher due to limited historical data.
- At start cycle 100 (late prediction): Errors significantly reduce, that is the model captures the degradation behaviours more accurately.

The overall results shows that our proposed hybrid CNN–LSTM framework efficiently models the battery degradation behaviours and provides long-term predictions, validating its suitability for real-world prognostics and health management applications.

V. CONCLUSION

A hybrid CNN–LSTM architecture for predicting the RUL. It is a single framework model that identifies useful features and understands how battery behaviour changes across cycles. The dataset used comes from the NASA Prognostics Center of Excellence Lithium-ion Battery Repository, the batteries in the dataset, B0005, B0006, B0007, and B0018 have been charged and discharged continuously over many cycles and they are known to have varying degradation characteristics. For this reason, these batteries were chosen for our analysis. It is being concluded from the observed results that the architecture is able to achieve low error rate values being evaluated in the terms of MAPE, RMSE, and MAE. The proposed model is able to perform well in both the early and late prediction stages and it can adapt to the batteries that degrade in different patterns. The results show that the proposed model can effectively understand battery degradation patterns. These models can also be used in Battery Management Systems (BMS) in real time. This helps improve predictive maintenance and increases the overall safety of the battery system. RUL estimation improves the reliability of the battery systems and helps manage the lifecycle of batteries in EVs. Future scope in this regard will be more focused towards enhancing the early stage predictions by exploring the methods like attention-based models and transformers.

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